

Question

Using water with an ^{18}O content of 0.85% to dissolve normal potassium carbonate and potassium bicarbonate, a mixed aqueous solution of K_2CO_3 and KHCO_3 was prepared. Under light conditions, this solution was supplied to *Chlorella*. After a period of time, the ^{18}O content in the carbonate, water, and released oxygen was measured to be 0.61%, 0.85%, and 0.86%, respectively. It is known that oxygen exists in nature in the forms of ^{18}O , ^{17}O , and ^{16}O . The atomic weight of oxygen is 15.9994, while the atomic weights of ^{16}O , ^{17}O , and ^{18}O are 15.9949, 16.9991, and 17.9992, respectively. The isotopic abundance of ^{16}O is 99.762%.

Which of the following statements is correct:

- A. At the start of the experiment, the ^{18}O content in the carbonate was 0.31%.
- B. The gas produced by *Chlorella* photosynthesis originates from bicarbonate.
- C. In the absence of *Chlorella*, the ^{18}O content of carbonate does not change.
- D. Adding a small amount of sodium hydroxide solution with a ^{18}O content of 0.85% to the carbonate solution accelerates the rate of change in the ^{18}O content of the carbonate.
- E. None of the above options are correct.

Answer

Correct Answer: **E**

Detailed Explanation

Based on the isotopic abundance, let the content of ^{18}O in the carbonate be x , then

$$15.9949 \times 99.762\% + 16.9991 \times (1 - x - 99.762) + 17.9992x = 15.9994$$

Solving gives $x = 0.0021 = 0.21\%$, so option A is incorrect.

CHECKPOINT

1 PTS

^{18}O content in carbonate is 0.21%

Since the ^{18}O content in the released oxygen matches that in water but not in bicarbonate, the oxygen produced by *Chlorella* photosynthesis originates from water, making option B incorrect.

CHECKPOINT

1 PTS

^{18}O content in released oxygen matches that in water

CHECKPOINT

1 PTS

Oxygen produced by Chlorella photosynthesis originates from water

Carbonate can hydrolyze to produce carbon dioxide, which then reacts with water or hydroxide to enable ^{18}O exchange between water and carbonate. Thus, even without Chlorella, the ^{18}O content in carbonate would change, making option C incorrect.

CHECKPOINT

1 PTS

Carbonate can hydrolyze to produce carbon dioxide, which then reacts with water or hydroxide to enable ^{18}O exchange between water and carbonate

After adding sodium hydroxide, the solution's pH increases, reducing the concentrations of bicarbonate and carbonic acid and slowing the reaction, making option D incorrect.

CHECKPOINT

1 PTS

After adding sodium hydroxide, the solution's pH increases, reducing the concentrations of bicarbonate and carbonic acid and slowing the reaction

In summary, options A-D are all incorrect, and option E is correct.